

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed **end-to-end platform architectures** for robotics systems, integrating **sensing, compute, connectivity, and motion** to create reusable building blocks, achieving **95% component reusability** across prototype iterations.
- Designed and executed **characterization, benchmarking, and test automation** for robotics prototypes, generating performance and robustness data that informed critical architecture choices and accelerated technology readiness by **20%**.
- Built and iterated on **rapid prototypes** encompassing electrical, mechanical, firmware, and software components to evaluate new robotics concepts and de-risk system feasibility, reducing initial concept validation time by **30%**.
- Authored comprehensive **design docs, integration guides, and internal enablement materials**, capturing system tradeoffs and recommended patterns to transfer learnings to product teams, improving cross-functional understanding by **40%**.
- Managed **production support and incident response** for enterprise applications, resolving P1/P2 incidents within SLA targets and reducing system downtime by **15%** through proactive monitoring using **Splunk** and **Dynatrace**.
- Automated data extraction and transformation tasks using **Python scripting** and **SQL** for scheduled reporting and ETL processes, improving data delivery efficiency by **25%** and reducing manual effort.

SKILLS

Robotics & Systems Engineering – Rapid Prototyping, End-to-End Platform Architectures, System Integration, Characterization, Benchmarking, Test Automation, Design Documentation, Firmware Development, Embedded Systems, ROS/ROS2

Software Development – Java Spring Boot, Python, REST APIs, Microservices, Flask, FastAPI, Docker, Git, CI/CD, Jenkins

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Linux, Shell Scripting

Data Analytics & BI – Power BI, DAX, Power Query, Tableau, R, RStudio, SQL (PostgreSQL, MySQL, Oracle), Data Modelling, Data Warehousing, Statistical Analysis

Observability & Support – Splunk, Dynatrace, CloudWatch, Incident Response, SLA Management, Production Support

Enterprise Systems – ERP Support, Business Systems Integration, Change Management, Jira, Confluence

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Autonomous Drone Navigation System (HCL Technologies)

- Developed a **Python-based navigation system** for autonomous drones, integrating real-time sensor data (simulated lidar and camera feeds) and implementing pathfinding algorithms to achieve **98% mission success rate** in simulated environments.
- Designed and built a **RESTful API service using FastAPI** to manage drone telemetry and command-and-control, enabling seamless communication between the ground station (UI) and the drone's onboard processing unit, reducing latency by **25%**.
- Implemented **robust error handling and logging** using **AWS CloudWatch** and **PostgreSQL** for data persistence, ensuring reliable operation and facilitating post-mission analysis of flight parameters and system performance.

Real-Time Transaction Monitoring Dashboard (Financial Services Client)

- Engineered a **real-time dashboard** for a financial-services client to monitor transaction flows and operational alerts, processing over **1 million transactions daily**.
- Developed automated anomaly detection rules and alerts using **Java Spring Boot** and **Python**, identifying and flagging failed or stuck payment messages, reducing manual intervention by **35%**.
- Collaborated with business analysts to define requirements for **transaction monitoring and reconciliation workflows**, ensuring adherence to regulatory and GDPR awareness in production environments.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)